

FinGPT

FinRobot

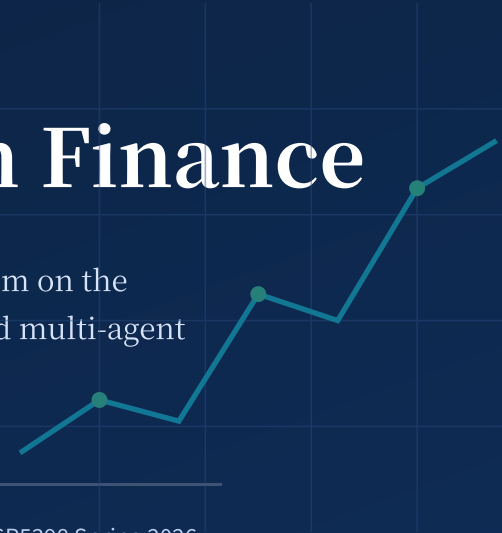
FinRL

FinRL-X

RESEARCH PROJECT • 2026

AI Agents, LLMs & Quantitative Trading in Finance

Design, implement, and evaluate an original financial AI system on the open-source AI4Finance ecosystem — from financial LLMs and multi-agent research to backtesting, paper trading, and deployment.

**Course** • STAT GR5380 Research Project 2026**Prerequisite reading** • STAT GR5398 Spring 2026 projects**Format** • Team project • GitHub repo + report + demo**Version** • v1.0 • July 2026**01** Overview & Research Platforms**02** Project Directions**03** Requirements & Deliverables**04** Evaluation & Research Standards**05** Timeline, Outcomes & Resources

01 OVERVIEW & RESEARCH PLATFORMS

Overview & Research Platforms

This year you build a complete **financial AI system**, not a single model: combining financial LLMs, AI agents, reinforcement learning, data engineering, portfolio & risk management, and backtesting → paper trading → deployment. Use one platform, or integrate several.

Platform	What it is	Example research topics
FinGPT	Financial large language models	News & social sentiment • instruction tuning & fine-tuning • RAG for financial research • earnings-call & SEC-filing analysis • financial QA • regime detection • LLM benchmarking
FinRobot	AI agents for financial research	Multi-agent equity research • statement & earnings analysis • AI research reports • DCF / DDM / comps / risk agents • bull–bear–judge debate • investment committee • agent evaluation
FinRL	Deep RL for quantitative finance	RL stock trading • portfolio allocation • risk-aware RL • multi-asset trading • reward design • environment construction • ablations • robustness across regimes
FinRL-X	AI-native trading infrastructure	Modular strategies • ML/LLM signals • signal-to-portfolio • portfolio optimization • risk control • backtest & paper-trading integration • monitoring & attribution

One workflow, four entry points: FinGPT perceives → FinRobot reasons → FinRL learns to decide → FinRL-X deploys.

Two rules that apply to every project

Language-side (FinGPT / FinRobot)

Go beyond calling a general LLM API. Separate **deterministic calculations** from **LLM interpretations**; support every financial conclusion with traceable data, evidence, and reproducible calculations.

Trading-side (FinRL / FinRL-X)

Use disjoint train / validation / test periods — never evaluate on data used during development. Model transaction costs, slippage, turnover, and risk exposure; audit for look-ahead bias.

#FinancialLLM

#Agent

#RAG

#QuantTrading

#ReinforcementLearning

#OpenSource

02 PROJECT DIRECTIONS

Project Directions

Pick the financial question first, then choose platforms. The strongest projects connect at least two platforms. Six exemplar directions:

① News-to-Trading System

FinGPT extracts sentiment / events from news → convert into portfolio decisions → evaluate in FinRL / FinRL-X.

② AI Equity Research Agent

FinRobot collects company data, analyzes statements, runs valuation and risk, and produces an evidence-grounded research report.

③ Regime & Risk-Control Agent

LLM/ML system detects changing market conditions and dynamically adjusts exposure, sector allocation, or risk limits.

④ Multi-Agent Investment Committee

Fundamental, technical, sentiment, risk, bull & bear agents; a judge / PM agent aggregates them into a decision.

⑤ Research-to-Paper-Trading Pipeline

Train a strategy in FinRL, deploy in FinRL-X, and analyze gaps among backtest, simulated execution, and broker paper trading.

⑥ Financial Agent Evaluation Framework

Benchmarks for factual & numerical correctness, evidence quality, stability, tool use, and financial reasoning.

Scoping advice

- **One clear contribution:** a new agent, a better signal, a stronger benchmark, or a tighter pipeline.
- **Reserve the last third of the semester** for evaluation, ablations, and reproducibility — not new features.
- **Review last semester's projects** (STAT GR5398 Spring 2026) before finalizing your proposal.

GRADING PRINCIPLE

A technically complex implementation will **not** receive a high evaluation unless it addresses a meaningful financial problem and is supported by rigorous experiments.

03 REQUIREMENTS & DELIVERABLES

Requirements & Deliverables

Your final report must cover seven components; your submission must include eight artifacts.

#	Component	What to cover
1	Research Problem	Financial problem • users / scenario • importance • limits of existing approaches • your research question.
2	Literature & System Review	Papers, models, ML/RL methods, LLM & agent architectures, open-source implementations — and how you differ.
3	Data	Sources • frequency & coverage • cleaning • features • train/val/test periods • licensing • survivorship / selection / look-ahead biases.
4	Methodology & Architecture	Models & algorithms • agent roles • tools & interfaces • financial assumptions • evaluation design • architecture diagram required .
5	Working Prototype	Executable system on FinGPT / FinRobot / FinRL / FinRL-X (or an integration) — not a conceptual design.
6	Empirical Evaluation	Baselines + metrics; analyze risk, turnover, costs, robustness, out-of-sample — never return alone.
7	Reproducibility & Risk	Limitations • hallucination • leakage • overfitting • biases • regime dependency • execution assumptions • responsible use.

Final deliverables (D1–D8)

D1 • GitHub repository (public or instructor-accessible)	D2 • Complete README with install & run instructions
D3 • Source code and configuration files	D4 • Data-processing and experiment scripts
D5 • Final research report	D6 • Project presentation
D7 • Live or recorded system demonstration	D8 • Results & reproducibility documentation

THE REPOSITORY STANDARD

Another student or researcher must be able to **understand, install, run, and evaluate** your project from the repo alone.

Evaluation & Research Standards

A strategy is never judged by cumulative return alone. Pick metrics from every relevant family and compare against a credible baseline, out of sample.

Dimension	Metrics	Applies to
Prediction quality	Accuracy, precision, recall, F1, calibration	FinGPT signals • ML classifiers
Factuality & grounding	Factual & numerical correctness, retrieval / citation quality	FinGPT • FinRobot reports
Return & risk	Return, annualized return, Sharpe, Sortino, max drawdown, volatility, VaR, benchmark-relative	FinRL • FinRL-X strategies
Implementability	Turnover, transaction costs, execution assumptions	All trading systems
Agent capability	Task-completion rate, tool-use accuracy, stability	FinRobot workflows
Efficiency	Runtime, latency, computational cost	All systems

Grading criteria

Research originality • financial relevance • technical quality • experimental rigor • correct use of financial data • system completeness • reproducibility • quality of analysis • clarity of report & presentation • potential open-source contribution.

Common failure modes to audit before submitting

Look-ahead bias

Features use information unavailable at decision time → use point-in-time data.

Data leakage

Test data touched during development → freeze splits before modeling.

Survivorship bias

Universe contains only surviving firms → keep delisted assets.

Overfitting

Backtest Sharpe collapses out of sample → ablations & walk-forward tests.

Hallucination

LLM asserts unsupported facts → retrieval with citations; recompute numbers.

Execution fantasy

Fills at impossible prices → explicit cost / slippage model; paper-trade check.

RESPONSIBLE USE

Course systems are research prototypes — never investment advice, never real capital. Respect data licenses; report limitations and negative results honestly.

05 TIMELINE, OUTCOMES & RESOURCES

Timeline, Outcomes & Resources

Assignments & final report

The semester is structured as **three assignments plus one final report**. Each assignment is a self-contained milestone; the final report integrates and refines the three assignments into one complete research paper.

- ASSIGNMENT 1 • WEEKS 1–4**
Research proposal
Research problem & question • literature & system review • data plan • platform selection → becomes report components 1–3.
- ASSIGNMENT 2 • WEEKS 5–9**
Working prototype
End-to-end system on the chosen platform(s) with a simple baseline • methodology & architecture diagram • runs from a clean clone → becomes report components 4–5.
- ASSIGNMENT 3 • WEEKS 10–12**
Empirical evaluation
Baselines, ablations & out-of-sample tests • full metric suite • risk & reproducibility discussion → becomes report components 6–7.
- FINAL REPORT • WEEKS 13–14**
Integrate Assignments 1–3 into the final submission
Merge and polish the three assignments into one coherent research report, plus presentation, demo, and the complete repository (deliverables D1–D8).

Where strong projects go next

A merged module in FinGPT / FinRobot / FinRL / FinRL-X • a new open-source financial AI module • a technical report or paper • a conference / workshop submission • an AI4Finance community presentation • continued collaboration with maintainers.

Resources

Resource	Link
Previous course projects	github.com/AI4Finance-Foundation/STAT-GR5398-Spring-2026
FinGPT	github.com/AI4Finance-Foundation/FinGPT
FinRobot	github.com/AI4Finance-Foundation/FinRobot
FinRL	github.com/AI4Finance-Foundation/FinRL
FinRL-X	github.com/AI4Finance-Foundation/FinRL-Trading

AI4Finance Foundation*"Democratizing AI for Finance through Open Source."*GR5380 Research Project 2026
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